GTU School of Engineering and Technology

2-Week Summer Internship on "From Digital Design to Silicon Chip: RTL to GDS-II"

in Collaboration with eInfochips - An Arrow Company, Ahmedabad

Schedule of the Internship

From Digital Design to Silicon Chip: RTL to GDS-II

1st Week

Date	06 July 2026	07 July 2026	08 July 2026	09 July 2026	10 July 2026
11AM-1PM	ASIC Flow & Design for Testability Fundamentals and Scan Architecture (Ila vaghela)	Embedded Deterministic Test and ATPG Methodologies (Ila Vaghela)	Fault Simulation Techniques and Industrial Case Studies (Ila Vaghela)	Semiconductor Design Methodology and RTL Design Principles (Maitri)	Verilog HDL Modeling and Digital Design Implementation (Maitri)
Break 1:00 PM-2:00PM					
2PM-4PM	Hands on Session (Ila vaghela)	Hands on Session (Ila vaghela)	Hands on Session (Ila vaghela)	Hands on Session (Maitri)	Hands on Session (Maitri)
Break 4:00PM-4.30PM					
4.30PM-5PM	QnA Session (Ila vaghela)	QnA Session (Ila vaghela)	QnA Session (Ila vaghela)	QnA Session (Maitri)	QnA Session (Maitri)

2nd Week

	13 July 2026	14 July 2026	15 July 2026	16 July 2026	17 July 2026
11:00 AM-01:00PM	Logic Synthesis Methodologies and RTL-to-Gates Flow (Chandni Patel / Rutvik)	Placement Techniques and Physical Design Fundamentals (Chandni Patel / Rutvik)	Physical Design Sign-Off Analysis and Case Studies (Chandni Patel / Rutvik)	Verification Methodology Fundamentals and SystemVerilog Constructs (Ashish)	Reusable Verification Components and Advanced Verification Techniques (Ashish)
Break 1:00 PM-2:00PM					
02:00PM-04:00PM	Hands on Session (Chandni Patel / Rutvik)	Hands on Session (Chandni Patel / Rutvik)	Hands on Session (Chandni Patel / Rutvik)	Hands on Session (Ashish)	Hands on Session (Ashish)
Break 1:00 PM-2:00PM					
4.30PM-05:00PM	QnA Session (Chandni Patel / Rutvik)	QnA Session (Chandni Patel / Rutvik)	QnA Session (Chandni Patel / Rutvik)	QnA Session (Ashish)	QnA Session (Ashish)

GUJARAT TECHNOLOGICAL UNIVERSITY

(Established under Gujarat Act No. 20 of 2007)

ગુજરાત ટેકનોલોજીકલ યુનિવર્સિટી

(ગુજરાત અધિનિયમ ક્રમાંક: ૨૦/૨૦૦૨ દ્વારા સ્થાપિત)

Annexure 1

Enrollment no: 230130111040

STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: Krish Bhavsar

DIARY OF THE WEEK: Dt: 03/07/2026 **TO** 09/07/2026

DEPARTMENT: Electronics and Communication Engineering **SEM:** VII

NAME OF THE ORGANISATION: Indian Institute of Information Technology Vadodara

NAME OF THE PLANT/SECTION/DEPARTMENT: Dept. of Electronics and Communication Engg.

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: Dr. Kamal Kishor Jha

DESCRIPTION OF THE WORK DONE IN BRIEF

Project: VLSI and AI-based ECG Arrhythmia Classification using 1D and 2D Convolutional Neural Networks

1. Literature Study and Problem Formulation

- Studied the clinical background of cardiac arrhythmia and the morphology of the ECG waveform, along with the four-class labelling scheme used in the PhysioNet/CinC Challenge 2017 dataset: Normal (N), Atrial Fibrillation (A), Other rhythm (O) and Noisy (~).
- Reviewed literature on deep-learning-based ECG classification, covering both 1D-CNN architectures operating directly on the raw waveform and 2D-CNN architectures operating on time-frequency image representations of the signal.
- Configured the development environment with Python 3.11, TensorFlow/Keras, NumPy, SciPy and scikit-learn under WSL Ubuntu, together with the WFDB package for reading PhysioNet-format ECG records.

2. Dataset Preparation - PhysioNet/CinC Challenge 2017

- Acquired and organised the PhysioNet/CinC Challenge 2017 database of 8,528 single-lead ECG recordings labelled into four classes: Normal (N), Atrial Fibrillation (A), Other rhythm (O) and Noisy (~).
- Implemented the pre-processing pipeline: band-pass filtering and baseline-wander removal, amplitude normalisation, R-peak detection and fixed-length windowing.

3. 1D-CNN Baseline Implementation and Verification

- Implemented a baseline 1D-CNN of stacked convolution, batch-normalisation, ReLU and max-pooling blocks followed by global pooling and a soft-max layer, and trained it on the pre-processed waveforms for the N / A / O / ~ four-class classification task.
- Translated the trained model into nine hand-coded Verilog RTL modules (MAC unit, 1D convolution engine, max-pooling, global average pooling, dense layers, argmax classifier, top-level wrapper and an AXI4-Lite interface) and verified the design bit-perfect against a NumPy reference model.
- Identified and fixed a signed/unsigned comparison bug in the argmax classifier's part-select logic found during verification.

4. 2D-CNN Variant using Time-Frequency Representations

- Converted the same ECG recordings into 2D time-frequency representations - spectrograms, scalograms and DWT (discrete wavelet transform) maps - using STFT/CWT/DWT.
- Implemented an SE-ResNet architecture on these 2D representations for the same N / A / O / ~ classification task and compared it against the 1D-CNN baseline.
- Observed the 2D-CNN (SE-ResNet) reach a macro F1-score of approximately 0.66, against approximately 0.76 for the 1D-CNN baseline, and documented the trade-off between the two approaches.

5. Hardware Deployment on PYNQ-Z2

- Targeted the verified 1D-CNN accelerator for deployment on the Xilinx PYNQ-Z2 FPGA board and compiled the week's 1D- and 2D-CNN results into a consolidated report.

TOTAL HOURS: 42

SIGNATURE OF STUDENT

✦ **The above entries are correct and the grading of work done by Trainee is EXCELLENT / VERY GOOD / GOOD / FAIR / BELOW AVERAGE / POOR**

Signature of Faculty Mentor

Signature of officer-in-charge of Dept. / Section / Plant

Date:

Date:

✦ **Grading of Work, for trainee may be given depending upon your judgement about his Punctuality, Regularity, Sincerity, Interest taken, Work done etc.**

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(ગુજરાત અધિનિયમ ક્રમાંક: ૨૦/૨૦૦૨ દ્વારા સ્થાપિત)

Annexure 1

Enrollment no: 230130111040

STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: Krish Bhavsar

DIARY OF THE WEEK: Dt: 10/07/2026 **TO** 16/07/2026

DEPARTMENT: Electronics and Communication Engineering **SEM:** VII

NAME OF THE ORGANISATION: Indian Institute of Information Technology Vadodara

NAME OF THE PLANT/SECTION/DEPARTMENT: Dept. of Electronics and Communication Engg.

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: Dr. Kamal Kishor Jha

DESCRIPTION OF THE WORK DONE IN BRIEF

Project: GCN-based Pre-Routing Timing Slack Prediction (TimingGNN)

1. Transition to GCN-based Timing Prediction

- Moved from the ECG 1D-CNN accelerator work to a new research problem: predicting post-routing timing slack directly from a pre-routing netlist using a Graph Convolutional Network (GCN), to avoid the cost of full physical-design iterations.
- Studied reference literature on GNN-based EDA applications, in particular Guo et al. (DAC 2022) and Mirhoseini et al. (Nature 2021), to frame the problem and identify baseline approaches.

2. Data Pipeline Development

- Built a custom parser for DEF, Verilog and SDF files to extract circuit-graph structure and timing annotations, including support for the min::max SDF format.
- Implemented filtering of power and filler cells so that the constructed circuit graphs reflect only functionally relevant standard cells.
- Set up training data using the spm circuit (real design) together with synthetic circuits, reserving the usb circuit (unseen during training) as the generalisation benchmark.

3. Model Design and Experimentation

- Implemented the GCN-based timing slack predictor in PyTorch with configurable node-aggregation strategies: SUM+MAX pooling, attention-based aggregation and mean aggregation.
- Ran an ablation study comparing the aggregation strategies, which formed the core experimental contribution of the work, and evaluated model generalisation on the unseen usb circuit.
- Compiled the week's model-development and ablation results into a consolidated report.

TOTAL HOURS: 42

SIGNATURE OF STUDENT

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Signature of Faculty Mentor

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Indian Institute of Information Technology Vadodara

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Phone No.: 079 - 23977 501 | URL: www.iiitvadodara.ac.in

No: Date: 16/07/2026

TO WHOM IT MAY CONCERN

This is to certify that, Mr. Krish Bhavsar Enrollment No 230130111040, Student of Government Engineering College, Gandhinagar (Gujarat Technological University), B.E. Electronics and Communication Engineering, has successfully completed a two week Internship in the field of VLSI Machine Learning for Electronic Design Automation (EDA), based on Graph Convolutional Neural Networks (GCNN) for Circuit Netlist Representation and Design-Quality Prediction during the period of 03/07/2026 to 16/07/2026.

During the period of his internship program with us, he had been exposed to different modelling, experimentation and analysis processes, and was found sincere, dedicated and hardworking.

He carried out this work as a Research Intern at the Indian Institute of Information Technology, Vadodara (IIIT Vadodara), under the guidance of Dr. Kamal Kishor Jha. We wish him the very best for his future endeavours.

Authorised Signature Mentor Signature

Institute Head